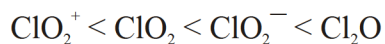
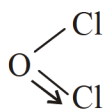
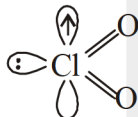
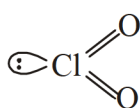


HINTS & SOLUTIONS

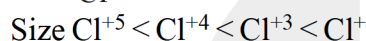
Q.1 (B)



$$\text{Bond length} \propto \frac{1}{\text{Bond order}}$$

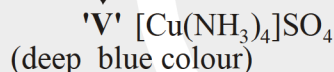
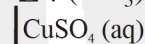
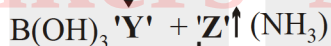
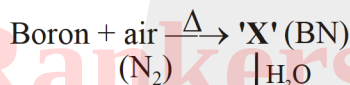


(back bonding)



more internuclear distance more will be bond length.

Q.2 (A)

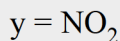
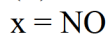


(deep blue colour)

Q.3 (D)

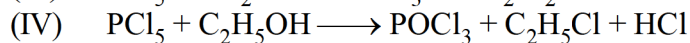
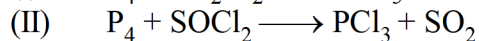
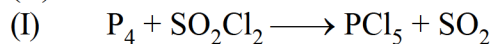
Q.4 (D)

Q.5 (B)



Q.6 (A)

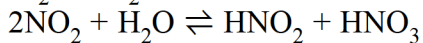
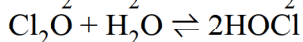
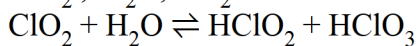
Q.7 (D)



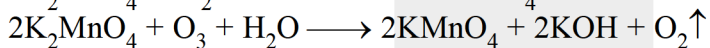
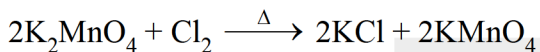
Q.8 (A)

Q.9 (B)

ClO_2 , Cl_2O , NO_2 are acidic.



Q.10 (C)

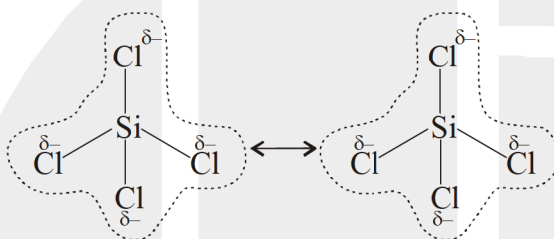


Q.11 (A)

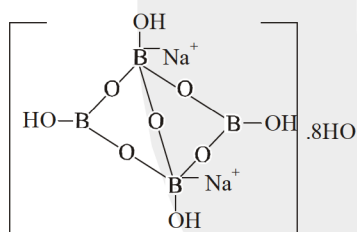
Red P does not produce phosphine on heating with NaOH

Q.12 (A)

Boiling point also depends upon difference in electronegativity between bonded atom. Due to more ΔE_n , more -ve charge density on bonded atom is created which repel each other resulting low boiling point.



Q.13 (A)



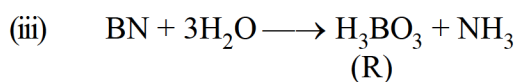
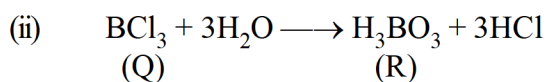
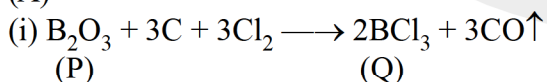
Q.14 (C)

As_2S_3 is yellow in colour.

Q.15 (C)

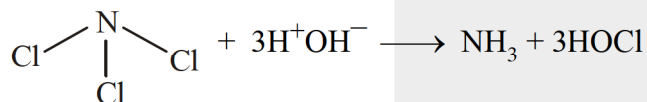
Q.16 (D)

Q.17 (A)

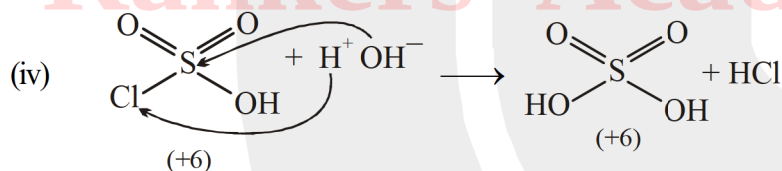
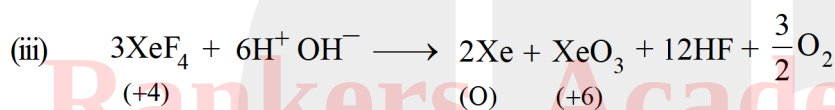
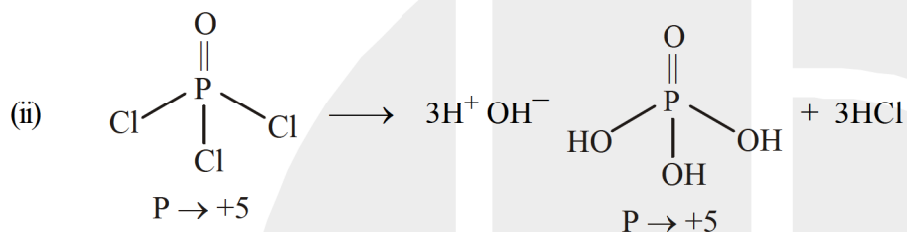
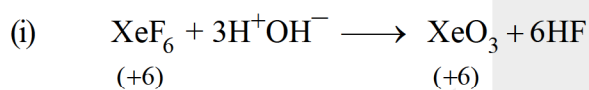


- (iv) $4\text{BCl}_3 + 3\text{LiAlH}_4 \longrightarrow 2\text{B}_2\text{H}_6 + 3\text{LiCl} + 3\text{AlCl}_3$
 (Q) (S)
- (v) $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \longrightarrow 2\text{H}_3\text{BO}_3 + 6\text{H}_2\uparrow$
 (S) (R)
- (vi) $\text{B}_2\text{H}_6 + 2\text{NaH} \longrightarrow 2\text{NaBH}_4$
 (S) (T)

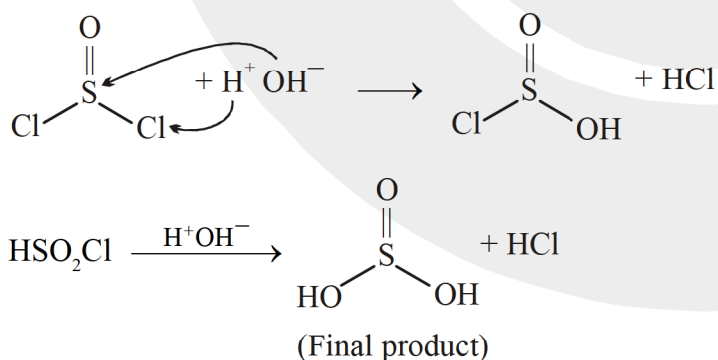
Q.18 (B)



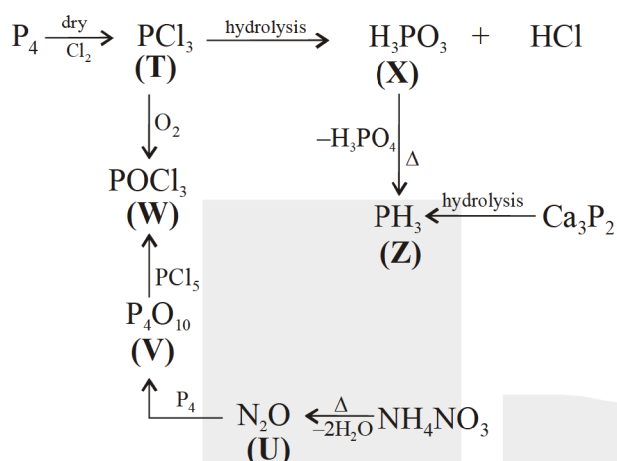
Q.19 (C)



Q.20 (A)



- Q.21 (ABC)
Q.22 (ACD)
Q.23 (ABCD)



- (i) (A) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \rightarrow 2\text{POCl}_3 + \text{SO}_2\text{Cl}_2 + 2\text{HCl}$
 (B) $\text{PCl}_5 + \text{SO}_2 \rightarrow \text{POCl}_3 + \text{SOCl}_2$
 (C) $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow \text{POCl}_3 + 2\text{HCl}$
 (ii) (A) $4\text{HI}_3\text{PO}_3 \xrightarrow{\Delta} 3\text{HI}_3\text{PO}_4 + \text{PII}_3$
 (B) $\text{NH}_4\text{NO}_3 \xrightarrow{\Delta} \text{N}_2\text{O} + 2\text{H}_2\text{O}$
 (C) N_2O supports combustion
 (D) $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 3\text{NaH}_2\text{PO}_2 + \text{PH}_3$
 (iii) $\text{N}_2\text{O} \rightarrow \text{N}_2 + [\text{O}]$

Q.24 (A)

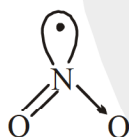
Q.25 (D)

- (i) NO_2 , N_2O_3 , N_2O_5 are acidic oxides
 NO : neutral oxide

N_2O neutral oxide

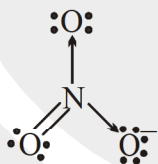
and CO neutral oxide

(ii)



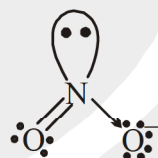
Total electrons = 23

sp^2 , Bent shape



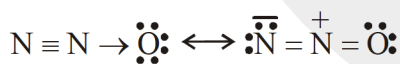
Total electrons = 32

sp^2 , Trigonal planar



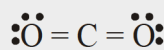
Total electrons = 24

sp^2 , Bent



Total electrons = 22

and



Total electrons = 22

are isoelectronic

& Linear in shape

Q.26 (D)

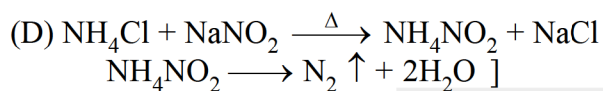
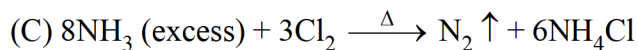
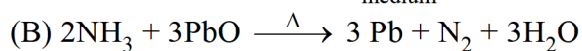
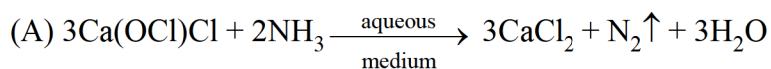
Q.27 (B)

Q.28 (ABD)

Q.29 (ABC)

- (D) Electrical conductivity of graphite is due to presence of free e^- .

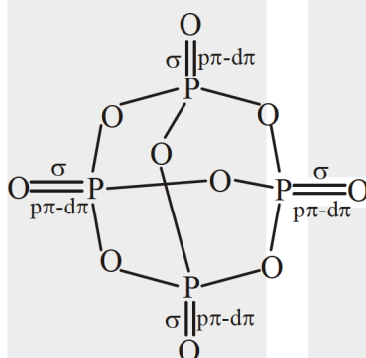
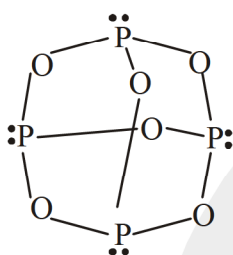
Q.30 (ABCD)



Q.31 (ABC)

Due to $p\pi-d\pi$ and / or $p\pi-p\pi$ back bonding.

Q.32 (ACD)

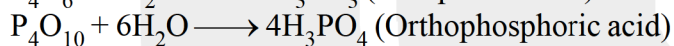


Both have cage like structure.

All six P—O—P bond in P_4O_6 are equivalent.

Similarly in P_4O_{10} all six P—O—P bonds are equivalent.

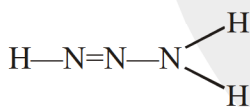
P_4O_6 has no $p\pi-d\pi$ bonds



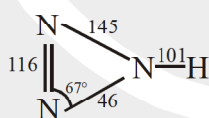
Q.33 (BC)

Q.34 (ABCD)

Q.35 (AB)



Triazene (N_3H_3)



Cyclotriazene (N_3H)

Q.36 (ACD)

In B_2H_6

(A) B—H

Terminal bond

2C—2 es^- bond

less strength

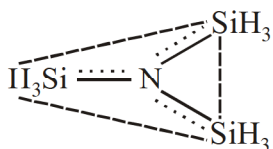
B—H...B

Bridge bond

3C—3 es^- bond

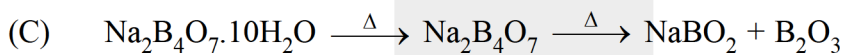
more strength

(B) Trisilylamine

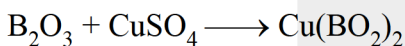


* Maximum 7 atoms are present in one plane.

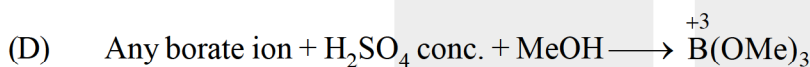
* N has sp^2 , triangular planar



Borax

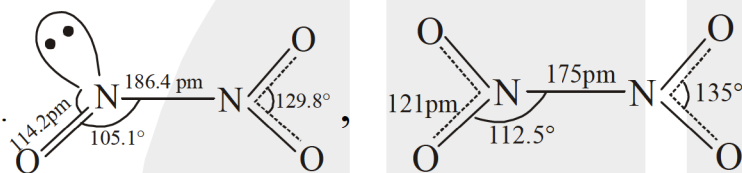


Blue bead in
oxidising flame



Give green edge flame.

Q.37 (ABCD)



Q.38 (A) PS (B) QS (C) QST (D) RS

Q.39 (0010)

(i) Due to decrease in electronegativity.

(ii) Due to middle row anomaly i.e. in the 4th period elements, effective nuclear charge is high due to poor shielding of d-electrons.

(iii) Due to increase in resonance stabilization.

(iv) Due to increase in bond length.

(v) Due to middle row anomaly.

(viii) Due to the increase in covalent character.

(ix) Due to the increasing stability of the lower species to which these are reduced

(x) Due to increase in Δ_0

Q.40 (2)

